

censtrunc

Censored and Truncated Normal Regression with Arbitrary Thresholds

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Package: `censtrunc` v0.1.0

Abstract

This report documents `censtrunc`, a Python package for maximum-likelihood estimation of censored and truncated normal regression models with **arbitrary left and right thresholds**. The package generalises the classical Type-1 Tobit model (Tobin, 1958) and the truncated normal regression to two-sided limits, while exposing a scikit-learn-style API and a statsmodels-style results summary. Estimation uses Olsen's (1978) globally concave reparameterisation for numerical reliability. The report presents the statistical model, then three worked examples: two-sided censoring on simulated data, the classical left-censored Tobit on the Fair (1978) affairs dataset, and a truncated regression.

1. Statistical Model

1.1 Latent regression

All models share the latent linear specification with normal errors:

$$Y^* = X'\beta + e, \quad e \mid X \sim \mathcal{N}(0, \sigma^2).$$

What differs is the rule linking the latent Y^* to the observed data.

1.2 Censored model

Values outside $[L, R]$ are **clipped** to the nearest threshold:

$$Y = \min(R, \max(L, Y^*)).$$

The log-likelihood combines the probability mass at each threshold with the normal density on the interior. With $\alpha_L = (L - X'\beta)/\sigma$ and $\alpha_R = (R - X'\beta)/\sigma$,

$$\ell(\beta, \sigma) = \sum_{Y_i=L} \log \Phi(\alpha_{L,i}) + \sum_{Y_i=R} \log [1 - \Phi(\alpha_{R,i})] + \sum_{L < Y_i < R} [\log \phi(\alpha_i) - \log \sigma],$$

where ϕ, Φ are the standard normal pdf and cdf. The classical Tobit is the special case $L = 0, R = +\infty$.

1.3 Olsen's reparameterisation

Optimisation is carried out in $(\gamma, \nu) = (\beta/\sigma, 1/\sigma)$. Olsen (1978) showed the resulting log-likelihood is **globally concave**, so any gradient-based optimiser converges to the unique maximum. After convergence the package back-transforms to (β, σ) and computes standard errors from the observed information matrix.

1.4 Truncated model

In a truncated sample, observations outside (L, R) are **absent** entirely. The density is the normal density renormalised by the probability of falling inside the interval:

$$f(y_i | L < Y_i^* < R) = \frac{\sigma^{-1} \phi((y_i - X_i' \beta)/\sigma)}{\Phi(\alpha_{R,i}) - \Phi(\alpha_{L,i})}.$$

1.5 Conditional means and marginal effects

Three conditional means are reported (Hansen, 2022, ch. 27):

- **Latent:** $\mathbb{E}[Y^* | X] = X' \beta$
- **Censored:** $\mathbb{E}[Y | X]$, accounting for the threshold mass
- **Truncated:** $\mathbb{E}[Y | X, L < Y < R]$

Marginal effects are available as **AME** (average over the sample) and **MEM** (evaluated at the mean regressor), each with delta-method standard errors. The censored marginal effect on $\mathbb{E}[Y | X]$ equals $\beta_j \cdot [\Phi(\alpha_R) - \Phi(\alpha_L)]$ — the latent slope shrunk by the probability of being interior.

1.6 Inference

Every fitted model carries an overall likelihood-ratio test against the intercept-only null model and McFadden's pseudo- R^2 . Arbitrary nested models can be compared with `lr_test`, which returns $LR = 2(\ell_{\text{full}} - \ell_{\text{restricted}}) \sim \chi_q^2$.

2. Worked Examples

The remainder of this report reproduces three example notebooks. Each is fully executed; the code and its output appear inline.

2.1 Example 1 — Two-Sided Censored Regression

```
In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from censtrunc import CensoredRegression, lr_test

rng = np.random.default_rng(2026)
n = 3000
```

Data-generating process

$$Y_i^* = 1.0 + 0.5 X_{1i} - 0.3 X_{2i} + 0.2 X_{3i} + e_i, \quad e_i \sim \mathcal{N}(0, 1).$$

The observation rule clips values below $L = 0$ and above $R = 2.5$, producing a realistic two-sided censoring pattern (think 'satisfaction score on a 0-2.5 scale').

```
In [2]: X = rng.normal(size=(n, 3))
beta_true = np.array([1.0, 0.5, -0.3, 0.2])
sigma_true = 1.0
y_star = beta_true[0] + X @ beta_true[1:] + rng.normal(scale=sigma_true, size=n)
L, R = 0.0, 2.5
y = np.clip(y_star, L, R)

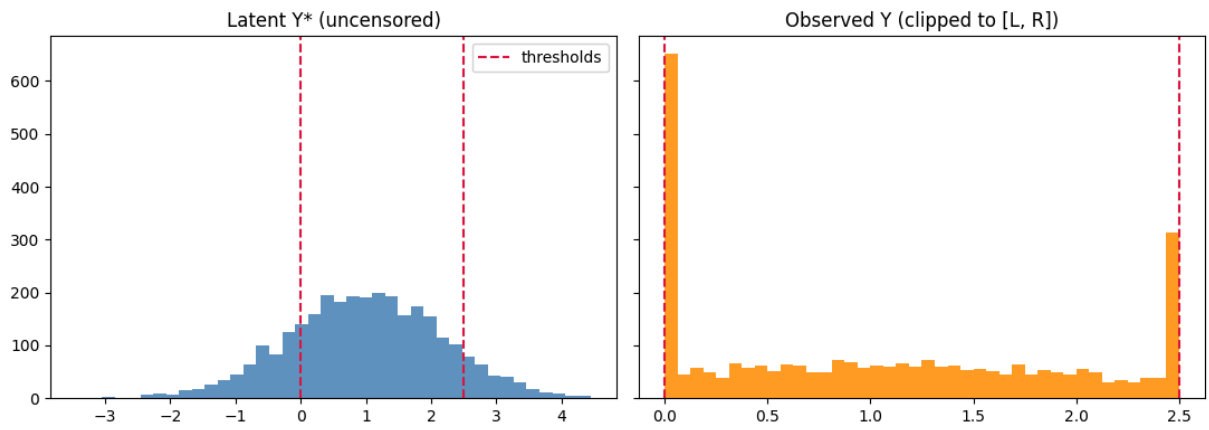
print(f'Left-censored:   {np.mean(y == L):.1%}')
print(f'Right-censored:  {np.mean(y == R):.1%}')
print(f'Interior:         {np.mean((y > L) & (y < R)):.1%}')
```

```
Left-censored:   20.0%
Right-censored:  9.8%
Interior:        70.2%
```

Distribution of the latent and observed variables

The pile-ups at L and R in the observed distribution are the hallmark of two-sided censoring.

```
In [3]: fig, axes = plt.subplots(1, 2, figsize=(11, 4), sharey=True)
axes[0].hist(y_star, bins=40, color='steelblue', alpha=0.85)
axes[0].set_title('Latent Y* (uncensored)')
axes[0].axvline(L, color='crimson', linestyle='--', label='thresholds')
axes[0].axvline(R, color='crimson', linestyle='--')
axes[0].legend()
axes[1].hist(y, bins=40, color='darkorange', alpha=0.85)
axes[1].set_title('Observed Y (clipped to [L, R])')
axes[1].axvline(L, color='crimson', linestyle='--')
axes[1].axvline(R, color='crimson', linestyle='--')
plt.tight_layout(); plt.show()
```



Fit the censored regression

The package's default optimisation uses Olsen's reparameterisation, which makes the log-likelihood globally concave and guarantees convergence of gradient-based optimisers (Olsen 1978).

```
In [4]: model = CensoredRegression(left=L, right=R).fit(
        X, y, feature_names=['x1', 'x2', 'x3']
        )
print(model.summary())
```

Censored Regression Results						
Dep. Variable:	y	No. Observations:	3000			
Model:	CensoredRegression	Df Model:	4			
Method:	MLE	Df Residuals:	2995			
Date:	Fri, 29 May 2026	Log-Likelihood:	-3831.9928			
Time:	22:22:28	LL-Null:	-4286.2006			
AIC:	7673.9855	LLR p-value:	1.323e-196			
BIC:	7704.0174	Pseudo R-squ.:	0.1060			
Left threshold:	0	Right threshold:	2.5			
	coef	std err	z	P> z	[0.025	0.975]
sigma	0.9959	0.0165	60.3216	0.0000	0.9635	1.0282
const	0.9963	0.0190	52.4873	0.0000	0.9591	1.0336
x1	0.5126	0.0195	26.2589	0.0000	0.4744	0.5509
x2	-0.2565	0.0188	-13.6315	0.0000	-0.2934	-0.2196
x3	0.2030	0.0192	10.5893	0.0000	0.1655	0.2406
Left-censored observations: 601 (20.03%)						
Right-censored observations: 293 (9.77%)						
Uncensored observations: 2106 (70.20%)						
Optimiser converged: True						

Compare against OLS

OLS on the censored sample is biased (Greene 1981): the slope coefficients are shrunk roughly in proportion to the censoring probability. Our Tobit-style MLE is consistent.

```
In [5]: X_with_int = np.column_stack([np.ones(n), X])
ols_beta, *_ = np.linalg.lstsq(X_with_int, y, rcond=None)

comparison = pd.DataFrame({
    'true': beta_true,
    'OLS (biased)': ols_beta,
```

```

'Censored MLE': model.coef_,
}, index=['const', 'x1', 'x2', 'x3'])
comparison['OLS bias'] = comparison['OLS (biased)'] - comparison['true']
comparison['MLE bias'] = comparison['Censored MLE'] - comparison['true']
comparison.round(4)

```

Out [5]:

	true	OLS (biased)	Censored MLE	OLS bias	MLE bias
const	1.0	1.0706	0.9963	0.0706	-0.0037
x1	0.5	0.3648	0.5126	-0.1352	0.0126
x2	-0.3	-0.1794	-0.2565	0.1206	0.0435
x3	0.2	0.1460	0.2030	-0.0540	0.0030

Predictions: the three flavours

- **latent** : $\mathbb{E}[Y^*|X] = X'\beta$
- **censored** : $\mathbb{E}[Y|X]$ — accounts for the threshold mass
- **truncated** : $\mathbb{E}[Y|X, L < Y < R]$ — conditional on being interior

In [6]:

```

X_show = X[:6]
preds = pd.DataFrame({
    'latent': model.predict(X_show, kind='latent'),
    'censored': model.predict(X_show, kind='censored'),
    'truncated': model.predict(X_show, kind='truncated'),
    'observed': y[:6],
})
preds.round(3)

```

Out [6]:

	latent	censored	truncated	observed
0	0.143	0.470	0.816	1.669
1	1.489	1.438	1.351	0.777
2	0.704	0.831	1.023	1.621
3	0.800	0.901	1.062	0.910
4	1.018	1.068	1.152	2.500
5	0.896	0.974	1.101	0.390

Probabilities of each region

predict_proba returns the conditional probability of each region given X . The three probabilities sum to 1 by construction.

In [7]:

```

proba = model.predict_proba(X_show)
pd.DataFrame(proba, index=range(6)).round(3)

```

Out [7]:

	left	interior	right
0	0.443	0.548	0.009
1	0.067	0.778	0.155
2	0.240	0.725	0.036
3	0.211	0.745	0.044
4	0.153	0.778	0.068
5	0.184	0.762	0.054

Marginal effects via `get_margeff`

The marginal-effects API mirrors statsmodels' `get_margeff` : one method, with options for *where* to evaluate (`at`) and *what* to report (`method`). The summary prints in the familiar statsmodels layout.

```
In [8]: # AME = average over the sample (at='overall'); statsmodels-style summary
print(model.get_margeff(at='overall', method='dydx', kind='censored').summary())
```

Censored Regression Marginal Effects						
Dep. Variable:	y					
Method:	dydx					
At:	overall					
	dy/dx	std err	z	P> z	[0.025	0.975]
x1	0.3600	0.012	30.227	0.000	0.337	0.383
x2	-0.1801	0.013	-14.113	0.000	-0.205	-0.155
x3	0.1426	0.013	10.805	0.000	0.117	0.168

`at='overall'` is the AME; `at='mean'` is the marginal effect at the mean (MEM).

Below we compare the censored marginal effect across evaluation points and against the latent effect (which equals β exactly).

```
In [9]: import pandas as pd
summary = pd.DataFrame({
    'latent (=beta)': model.get_margeff(at='overall', kind='latent').margeff,
    'censored AME': model.get_margeff(at='overall', kind='censored').margeff,
    'censored MEM': model.get_margeff(at='mean', kind='censored').margeff,
    'truncated AME': model.get_margeff(at='overall', kind='truncated').margeff,
}, index=['x1', 'x2', 'x3'])
summary.round(4)
```

Out [9]:

	latent (=beta)	censored AME	censored MEM	truncated AME
x1	0.5126	0.3600	0.3973	0.2005
x2	-0.2565	-0.1801	-0.1988	-0.1003
x3	0.2030	0.1426	0.1573	0.0794

`method` switches between the derivative and elasticities: `dydx` (derivative), `eyex` (elasticity), `dyex / eydx` (semi-elasticities). Discrete regressors can be handled with `dummy=True` (a 0->1 difference rather than a derivative).

```
In [10]: elas = model.get_margeff(at='overall', method='eyex', kind='censored')
         elas.summary_frame().round(4)
```

```
Out[10]:
```

	eyex	std err	z	P> z	[0.025	0.975]
x1	-0.1238	0.0093	-13.3389	0.0	-0.1420	-0.1056
x2	-0.0317	0.0045	-7.0004	0.0	-0.0406	-0.0228
x3	-0.0193	0.0035	-5.4717	0.0	-0.0263	-0.0124

Likelihood-ratio test

We test whether `x3` is jointly redundant by fitting a restricted model that omits it, then comparing log-likelihoods.

```
In [11]: restricted = CensoredRegression(left=L, right=R).fit(X[:, :2], y)
         res = lr_test(model, restricted)
         print(res)
```

```
Likelihood-ratio test
LR statistic : 111.3544
df           : 1
p-value      : 4.949e-26
log-lik full : -3831.9928
log-lik rest.: -3887.6700
n            : 3000
```

If the p-value is small, we reject the restriction (drop `x3`). Here `x3` has a true coefficient of 0.2 and a large sample, so the test correctly rejects.

2.2 Example 2 — Classical Tobit (Fair Affairs Data)

```
In [12]: import numpy as np
         import pandas as pd
         import statsmodels.api as sm
         from censtrunc import CensoredRegression

         data = sm.datasets.fair.load_pandas().data
         print('n =', len(data))
         print(f"share with affairs == 0: {(data['affairs'] == 0).mean():.1%}")
         data.head()
```

```
n = 6366
share with affairs == 0: 67.8%
```

```
Out [12]:
```

	rate_marriage	age	yrs_married	children	religious	educ	occupation	occupation_husb
0	3.0	32.0	9.0	3.0	3.0	17.0	2.0	
1	3.0	27.0	13.0	3.0	1.0	14.0	3.0	
2	4.0	22.0	2.5	0.0	1.0	16.0	3.0	
3	4.0	37.0	16.5	4.0	3.0	16.0	5.0	
4	5.0	27.0	9.0	1.0	1.0	14.0	3.0	

Specification

Following Fair (1978), we regress the count of affairs on the standard set of demographic and marital variables.

```
In [13]: covariates = ['rate_marriage', 'age', 'yrs_married', 'children',
                    'religious', 'educ', 'occupation', 'occupation_husb']
X = data[covariates]
y = data['affairs']
```

Fit

We tell `CensoredRegression` that the left threshold is zero; the right threshold is left unspecified.

```
In [14]: model = CensoredRegression(left=0.0).fit(X, y)
print(model.summary())
```

```
=====
                        Censored Regression Results
=====
Dep. Variable:          y                No. Observations:      6366
Model:                  CensoredRegression    Df Model:              9
Method:                  MLE                 Df Residuals:         6356
Date:                   Fri, 29 May 2026      Log-Likelihood:       -7804.3803
Time:                   22:22:29              LL-Null:              -8155.0240
AIC:                   15628.7605             LLR p-value:          3.780e-146
BIC:                   15696.3478             Pseudo R-squ.:        0.0430
Left threshold:         0                   Right threshold:      none
=====
```

	coef	std err	z	P> z	[0.025	0.975]
sigma	4.4990	0.0771	58.3463	0.0000	4.3478	4.6501
const	7.8354	0.7135	10.9818	0.0000	6.4370	9.2338
rate_marriage	-1.5313	0.0734	-20.8514	0.0000	-1.6752	-1.3873
age	-0.1051	0.0248	-4.2393	0.0000	-0.1537	-0.0565
yrs_married	0.1282	0.0264	4.8576	0.0000	0.0765	0.1799
children	-0.0276	0.0771	-0.3576	0.7206	-0.1788	0.1236
religious	-0.9434	0.0849	-11.1077	0.0000	-1.1099	-0.7769
educ	-0.0857	0.0376	-2.2772	0.0228	-0.1594	-0.0119
occupation	0.3125	0.0819	3.8150	0.0001	0.1520	0.4731
occupation_husb	0.0143	0.0555	0.2578	0.7966	-0.0944	0.1230

```
=====
Left-censored observations:      4313 (67.75%)
Right-censored observations:      0 ( 0.00%)
Uncensored observations:       2053 (32.25%)
Optimiser converged:            True
=====
```


Marginal effects

Because the response is left-censored at zero, the censored marginal effects are smaller in absolute value than the latent ones. The latent (uncensored) coefficients describe the propensity scale, while the censored marginal effects describe the *expected number* of affairs.

```
In [15]: ame_cens = model.ame(kind='censored').to_dataframe()
ame_lat = model.ame(kind='latent').to_dataframe()
pd.concat([
    ame_lat[['dy/dx']].rename(columns={'dy/dx': 'latent'}),
    ame_cens[['dy/dx', 'std err', 'P>|z|']].rename(columns={'dy/dx': 'censored E[Y]'}),
], axis=1).round(4)
```

```
Out[15]:
```

	latent	censored E[Y]	std err	P> z
rate_marriage	-1.5313	-0.4326	0.0213	0.0000
age	-0.1051	-0.0297	0.0070	0.0000
yrs_married	0.1282	0.0362	0.0074	0.0000
children	-0.0276	-0.0078	0.0218	0.7206
religious	-0.9434	-0.2665	0.0243	0.0000
educ	-0.0857	-0.0242	0.0106	0.0228
occupation	0.3125	0.0883	0.0232	0.0001
occupation_husb	0.0143	0.0040	0.0157	0.7966

Sanity check vs. OLS

Naive OLS on the censored sample understates the slope coefficients (Greene 1981). The Tobit MLE corrects this.

```
In [16]: X_int = sm.add_constant(X)
ols = sm.OLS(y, X_int).fit()
pd.DataFrame({
    'OLS coef': ols.params.values,
    'Tobit coef': model.coef_,
}, index=ols.params.index).round(4)
```

Out [16]:

	OLS coef	Tobit coef
const	3.6235	7.8354
rate_marriage	-0.4205	-1.5313
age	-0.0146	-0.1051
yrs_married	-0.0160	0.1282
children	-0.0171	-0.0276
religious	-0.2437	-0.9434
educ	-0.0174	-0.0857
occupation	0.0658	0.3125
occupation_husb	0.0040	0.0143

2.3 Example 3 — Truncated Regression

```
In [17]: import numpy as np
import pandas as pd
from censtrunc import TruncatedRegression

rng = np.random.default_rng(7)
n_population = 8000
```

Simulate a truncated dataset

We draw a large population, then keep only observations with $0 < Y^* < 2.5$ — about 60% of the population survives the double-sided truncation in this DGP.

```
In [18]: X_pop = rng.normal(size=(n_population, 2))
beta_true = np.array([1.0, 0.5, -0.3])
y_star = beta_true[0] + X_pop @ beta_true[1:] + rng.normal(scale=1.0, size=n_population)
L, R = 0.0, 2.5
mask = (y_star > L) & (y_star < R)
X = X_pop[mask]
y = y_star[mask]
print(f'Observed after truncation: {len(y)} / {n_population}')
```

Observed after truncation: 5741 / 8000

Fit the truncated regression

```
In [19]: model = TruncatedRegression(left=L, right=R).fit(
    X, y, feature_names=['x1', 'x2']
)
print(model.summary())
```

Truncated Regression Results						
Dep. Variable:	y	No. Observations:	5741			
Model:	TruncatedRegression	Df Model:	3			
Method:	MLE	Df Residuals:	5737			
Date:	Fri, 29 May 2026	Log-Likelihood:	-4830.9612			
Time:	22:22:29	LL-Null:	-5136.1192			
AIC:	9669.9225	LLR p-value:	2.962e-133			
BIC:	9696.5440	Pseudo R-squ.:	0.0594			
Left truncation:	0	Right truncation:	2.5			
	coef	std err	z	P> z	[0.025	0.975]
sigma	0.9944	0.0316	31.5070	0.0000	0.9326	1.0563
const	0.9946	0.0245	40.6756	0.0000	0.9467	1.0425
x1	0.4881	0.0335	14.5918	0.0000	0.4226	0.5537
x2	-0.3105	0.0268	-11.6040	0.0000	-0.3629	-0.2580
Optimiser converged:	True					

Comparison: OLS on the truncated sample is biased

Naive OLS on the truncated sample treats the data as if it came from an unrestricted population — leading to biased slope estimates, typically shrunk toward zero.

```
In [20]: X_int = np.column_stack([np.ones(len(y)), X])
ols_beta, *_ = np.linalg.lstsq(X_int, y, rcond=None)

comparison = pd.DataFrame({
    'true': beta_true,
    'OLS (biased)': ols_beta,
    'Truncated MLE': model.coef_,
}, index=['const', 'x1', 'x2'])
comparison['OLS bias'] = comparison['OLS (biased)'] - comparison['true']
comparison['Truncated MLE bias'] = comparison['Truncated MLE'] - comparison['true']
comparison.round(4)
```

```
Out[20]:
```

	true	OLS (biased)	Truncated MLE	OLS bias	Truncated MLE bias
const	1.0	1.1475	0.9946	0.1475	-0.0054
x1	0.5	0.1952	0.4881	-0.3048	-0.0119
x2	-0.3	-0.1241	-0.3105	0.1759	-0.0105

Predictions

Two prediction kinds are available for the truncated model:

- **latent** : $X'\hat{\beta}$ — the population mean
- **truncated** : $\mathbb{E}[Y|X, L < Y < R]$

```
In [21]: preds = pd.DataFrame({
    'observed': y[:6],
    'latent': model.predict(X[:6], kind='latent'),
    'truncated': model.predict(X[:6], kind='truncated'),
}).round(3)
preds
```

Out [21]:

	observed	latent	truncated
0	1.590	0.902	1.104
1	0.317	1.137	1.202
2	1.514	1.081	1.178
3	1.376	0.947	1.122
4	1.085	0.764	1.047
5	0.074	0.481	0.936

2.4 Example 4 — Validation against OLS and R

In [22]:

```
import numpy as np
import pandas as pd
import statsmodels.api as sm
from censtrunc import CensoredRegression, TruncatedRegression

rng = np.random.default_rng(0)
n = 1000
X = rng.normal(size=(n, 3))
y = 2.0 + 1.5*X[:,0] - 0.7*X[:,1] + 0.3*X[:,2] + rng.normal(scale=1.3, size=n)
```

No censoring $\Rightarrow$ OLS

We set `left=-1e6`, `right=1e6` so that no observation is censored. The censored log-likelihood then reduces to the Gaussian log-likelihood, whose maximiser is exactly the OLS coefficient vector.

In [23]:

```
ols = sm.OLS(y, sm.add_constant(X)).fit()
cens = CensoredRegression(left=-1e6, right=1e6).fit(X, y)
trunc = TruncatedRegression(left=-1e6, right=1e6).fit(X, y)

pd.DataFrame({
    'OLS': np.asarray(ols.params),
    'Censored MLE': cens.coef_,
    'Truncated MLE': trunc.coef_,
}, index=['const', 'x1', 'x2', 'x3']).round(6)
```

Out [23]:

	OLS	Censored MLE	Truncated MLE
const	2.046512	2.046482	2.046512
x1	1.434191	1.434221	1.434191
x2	-0.626598	-0.626603	-0.626598
x3	0.345980	0.345982	0.345980

The columns agree to several decimals. We can quantify the maximum discrepancy and confirm the scale parameter and log-likelihood match too (using the MLE scale

$$\hat{\sigma} = \sqrt{\text{SSR}/n}.$$

```
In [24]: ols_beta = np.asarray(ols.params)
print(f'max |beta_censored - beta_OLS| = {np.max(np.abs(cens.coef_ - ols_beta)):.2e}')
print(f'max |beta_truncated - beta_OLS| = {np.max(np.abs(trunc.coef_ - ols_beta)):.2e}')
print(f'|sigma_censored - sqrt(SSR/n)| = {abs(cens.sigma_ - np.sqrt(ols.ssr/n)):.2e}')
print(f'|loglik_censored - loglik_OLS| = {abs(cens.llf_ - ols.llf):.2e}')
```

```
max |beta_censored - beta_OLS| = 3.00e-05
max |beta_truncated - beta_OLS| = 4.44e-16
|sigma_censored - sqrt(SSR/n)| = 7.55e-07
|loglik_censored - loglik_OLS| = 5.90e-07
```

All discrepancies are at the level of the optimiser tolerance — the estimator behaves exactly as theory requires in the no-censoring limit, which is strong evidence the likelihood and its optimisation are implemented correctly.